

Assignment ITSQL10 - JDTF11

Q1: A retail company wants to reward its most valuable customers. Display the top 5 highest-value orders along with customer names.

Ans:

Query:-

```
select customername, orderamount from orders
inner join customers
on Customers.customerid=orders.customerid
order by orderamount desc limit 5;
```

Output:-

customername	orderamount
Vikram Patel	42000.00
Rahul Sharma	30000.00
Priya Mehta	27000.00
Amit Verma	25000.00
Rahul Sharma	22000.00

Q2: The marketing team wants to send promotional emails to customers whose names start with 'A', 'R', or 'S'. Write a query without using multiple OR conditions.

Ans:

Query:-

```
Select customername from customers
where left (customername, 1) in ("A","R","S");
```

Output:-

customername
Rahul Sharma
Arjun Nair
Sneha Gupta
Amit Verma
Rohan Singh
Simran Kaur
Anjali Jain

Q3: Management wants to identify customers who have placed orders worth more than ₹10,000 but whose order status is still Pending. Display:

- Customer Name
- Order ID
- Order Amount
- Status

Ans:

Query:-

Select customername,orderid,orderamount, status from customers
inner join orders
on customers.customerid=orders.CustomerID
where orderamount>10000 and status ="pending";

Output:-

	customername	orderid	orderamount	status
▶	Rohan Singh	5015	15500.00	Pending

Q4: The customer success team wants to identify customers who joined more than 365 days ago. Display the number of days each customer has been associated with the company.

Ans:

Query:-

select customername, timestampdiff(day,joindate,current_date()) as "No. of Days"
from customers
where "no. of days" > "365";

Output:-

customername	No. of Days
Rahul Sharma	1337
Priya Mehta	907
Arjun Nair	609
Sneha Gupta	1526
Kavya Reddy	759
Amit Verma	1043
Rohan Singh	848
Simran Kaur	589
Anjali Jain	1087
Vikram Patel	1353

Q5: The finance department wants to estimate future revenue.

Create a report showing:

- Order ID
- Current Amount
- 18% GST
- Total Amount After GST
- Expected Collection Date (15 days after order date)

Ans:

Query:-

```
SELECT orderid, orderamount AS current_amount, orderamount * 0.18 AS gst_18,  
orderamount * 1.18 AS total_amount_after_gst, DATE_ADD(orderdate, INTERVAL 15 DAY) AS  
expected_collection_date  
FROM orders;
```

Output:-

orderid	current_amount	gst_18	total_amount_after_gst	expected_collection_date
5001	12000.00	2160.0000	14160.0000	2025-01-20
5002	8500.00	1530.0000	10030.0000	2025-01-30
5003	22000.00	3960.0000	25960.0000	2025-02-25
5004	4500.00	810.0000	5310.0000	2025-03-02
5005	17500.00	3150.0000	20650.0000	2025-03-16
5006	9800.00	1764.0000	11564.0000	2025-03-25
5007	25000.00	4500.0000	29500.0000	2025-04-02
5008	14500.00	2610.0000	17110.0000	2025-04-17
5009	30000.00	5400.0000	35400.0000	2025-04-25
5010	6500.00	1170.0000	7670.0000	2025-04-30
5011	42000.00	7560.0000	49560.0000	2025-05-16
5012	18000.00	3240.0000	21240.0000	2025-05-23
5013	7000.00	1260.0000	8260.0000	2025-05-30
5014	27000.00	4860.0000	31860.0000	2025-06-04
5015	15500.00	2790.0000	18290.0000	2025-06-16

Q6: The operations team wants to review the customers ranked between 11th and 20th highest orders. Use LIMIT and OFFSET.

Ans:

Query:-

Select * from customers
inner join orders
on customers.customerid=orders.customerid
Order by orderamount desc limit 10, 10;

Output:-

CustomerID	CustomerName	City	JoinDate	OrderID	CustomerID	OrderDate	OrderAmount	Status
105	Ka			5	105	2025-03-10	9800.00	Pending
102	Priya Mehta	Mumbai	2024-03-20	5002	102	2025-01-15	8500.00	Pending
106	Amit Verma	Pune	2023-11-05	5013	106	2025-05-15	7000.00	Cancelled
109	Anjali Jain	Mumbai	2023-09-22	5010	109	2025-04-15	6500.00	Pending
104	Sneha Gupta	Delhi	2022-07-10	5004	104	2025-02-15	4500.00	Cancelled

Q7: An e-commerce company wants to identify customers who have never placed any order despite registering on the platform.

Display:

- Customer ID
- Customer Name
- Join Date
- Number of Days Since Registration

Ans:

Query:-

Select customers.customerid, customername, joindate,
timestampdiff(day,joindate,current_date()) as "No. of Days Since Registration" from customers
left join orders
on Customers.customerid = orders.customerid
where orderid is null;

Output:-

customerid	customername	joindate	No. of Days Since Registration
108	Simran Kaur	2025-02-01	589

Q8: The HR department wants to analyze organizational hierarchy.

Using a JOIN, display:

- Employee Name
- Manager Name
- Department

Ans:

Query:-

```
Select tb1.employeename, tb2.employeename as "Manager Name", Tb1.department
from employees as tb1
join employees as Tb2
on Tb1. ManagerID = Tb2. employeeid;
```

Output:-

employeename	Manager Name	department
Ka Karan Mehta	Rajesh Kumar	HR
Amit Gupta	Rajesh Kumar	IT
Neha Sharma	Rajesh Kumar	Sales
Vikas Patel	Neha Sharma	Sales
Priyanka Singh	Neha Sharma	Sales
Arjun Malhotra	Amit Gupta	IT
Anjali Verma	Amit Gupta	IT
Rohit Jain	Amit Gupta	IT
Sneha Kapoor	Karan Mehta	HR

Q9: The sales director wants a report of customers who placed orders within 30 days of joining the platform.

Display:

- Customer Name
- Join Date
- Order Date
- Days Taken To Place First Order

Ans:

Query:-

```
select customers.customername, joindate, orders.orderdate,
timestampdiff(day,joindate,orderdate) as "Days taken to place order"
from customers
inner join orders
on customers.customerid=orders.customerid
where "Days taken to place order" <="30";
```

Output:- "There is no customer who placed order within 30 days"

customername	joindate	orderdate	Days taken to place order
--------------	----------	-----------	---------------------------

Q10: Business Requirement

Generate a report showing:

- Customer Name
- Total Purchase Amount
- Spending Category
- Loyalty Status

Rules

Spending Category

- Premium → Total Purchases > ₹50,000
- Gold → ₹20,000 to ₹50,000
- Silver → Less than ₹20,000

Loyalty Status

- Joined before 2024 → Loyal Customer
- Otherwise → New Custom

Ans:

Query:-

Select customers.customername, orders.OrderAmount,
case

when orders.orderamount>50000 then "Premium"

when orders.orderamount>20000 then "Gold"

else "Silver"

End as "Spending Cateogary",

case

when customers.joindate<"2024-01-01" then "Loyal Customer"

else "New Customer"

end as "Loyalty Status"

from customers

inner join orders

on customers.CustomerID=orders.CustomerID;

Output:-

customername	OrderAmount	Spending Cateogary	Loyalty Status
Rahul Sharma	12000.00	Silver	Loyal Customer
Priya Mehta	8500.00	Silver	New Customer
Rahul Sharma	22000.00	Gold	Loyal Customer
Sneha Gupta	4500.00	Silver	Loyal Customer
Arjun Nair	17500.00	Silver	New Customer
Kavya Reddy	9800.00	Silver	New Customer
Amit Verma	25000.00	Gold	Loyal Customer
Rohan Singh	14500.00	Silver	New Customer
Rahul Sharma	30000.00	Gold	Loyal Customer
Anjali Jain	6500.00	Silver	Loyal Customer
Vikram Patel	42000.00	Gold	Loyal Customer
Sneha Gupta	18000.00	Silver	Loyal Customer
Amit Verma	7000.00	Silver	Loyal Customer
Priya Mehta	27000.00	Gold	New Customer
Rohan Singh	15500.00	Silver	New Customer

Q11: An online marketplace wants to find customers whose order amount is higher than the average order amount of all orders.

Display:

- Customer Name
- Order ID
- Order Amount

Ans:

Query:-

```
Select customers.customername, orders.OrderID, orders.OrderAmount
from customers
inner join orders
on customers.CustomerID=orders.CustomerID
where orderamount>(
select avg(orderamount) from orders);
```

Output:-

customername	OrderID	OrderAmount
Rahul Sharma	5009	30000.00
Rahul Sharma	5003	22000.00
Priya Mehta	5014	27000.00
Arjun Nair	5005	17500.00
Sneha Gupta	5012	18000.00
Amit Verma	5007	25000.00
Vikram Patel	5011	42000.00

Q12: The management team wants the 2nd to 6th highest orders placed by customers.

Display:

- Customer Name
- Order ID
- Order Amount

Ans:

Query:-

```
Select customers.customername, orders.OrderID, orders.OrderAmount
from customers
inner join orders
on customers.CustomerID=orders.CustomerID
order by OrderAmount desc limit 2,4;
```

Output:-

customername	OrderID	OrderAmount
Priya Mehta	5014	27000.00
Amit Verma	5007	25000.00
Rahul Sharma	5003	22000.00
Sneha Gupta	5012	18000.00